FIRST TOR OF THE GLUEING PAIR VANISHES AGAINST THE F-POWER TORSION QUOTIENT

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Allowed standard prerequisites (editorial).

- Tor groups commute with filtered colimits of modules. invoked at proof lines 25493-25495. Allows transferring the vanishing from each R[f^n]-quotient stage to the colimit R/R[f^infinity].

Source reference. Original lemma 0BNF: sources/stacks-chow/tags.txt, source lines 14663–14663. Cited at proof lines 25493-25495 to commute Tor^R_1 past the filtered colimit; the commutation is load-bearing inside the counted span.

Source reference. Original lemma 0BNZ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25472–25472. Cited at proof line 25495 to supply the vanishing of Tor_1(R', f^nR) at each colimit stage; used inside the counted span.

Lemma 0.2. *Let $(R \to R', f)$ be a glueing pair. Then $Tor_1^R(R', R/R[f^\infty]) = 0$.*

Proof. We have $R/R[f^\infty] = \operatorname{colim} R/R[f^n] = \operatorname{colim} f^n R$. As formation of Tor groups commutes with filtered colimits (Algebra, Lemma 0BNF) we may apply Lemma 0BNZ. □

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